

Table 3. Preservation of medical and general QA performance

Method	Qwen2.5 7B					GPT-6 Luna					Gemma 4 12B					Qwen3.8 27B OFF					Qwen3.8 27B ON				
	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K	MedQA	PubMedQA	Medbullets	MMLU	GSM8K
	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑	Acc. ↑	Acc. ↑	Acc. ↑	Acc. ↑	EM ↑
TF-IDF text gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Plain	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CoT	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Balanced instruction	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Premise-review CoT (2-step)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Always correct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
PreWoMe-style	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Extract + Verify	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Direct-gated response	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Premise-review gated response	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Probe-gated response	—	—	—	—	—	n/a	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GEPA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

All entries are unmeasured; — means no completed comparable result, not zero. n/a: native probe unavailable.
Acc.: accuracy (%); GSM8K EM: final-answer exact match (%). Datasets are reported separately.
Evaluate the frozen response methods from Table 2; do not retune them on these QA test sets.
Use identical question IDs and scoring rules across methods within each dataset and model.
Report performance relative to the same-model Plain baseline; Ours implementation remains unspecified.